MYSQL PROJECT

TOPIC: Myntra Product Analysis Using MySQL
MYSQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of a Myntra product dataset using MySQL. The dataset contains product-level information including brand name, product description, selling price, MRP, discount percentage, customer ratings, and number of ratings.

The objective of this project is to extract meaningful insights related to product pricing, brand performance, discounts, customer ratings, and product popularity using SQL techniques such as filtering, aggregation, GROUP BY, HAVING, and conditional expressions.

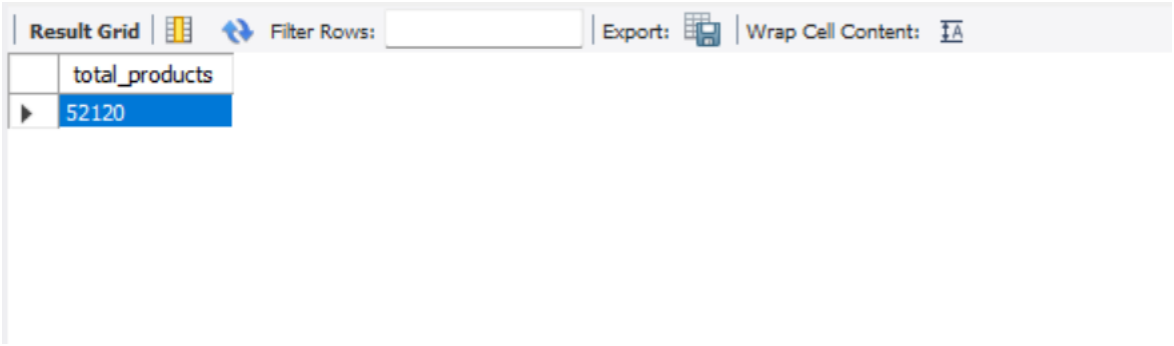
TOOL: MySQL Workbench

Submitted By – Meharin Ullas

Queries

1. **Find the total number of products.**

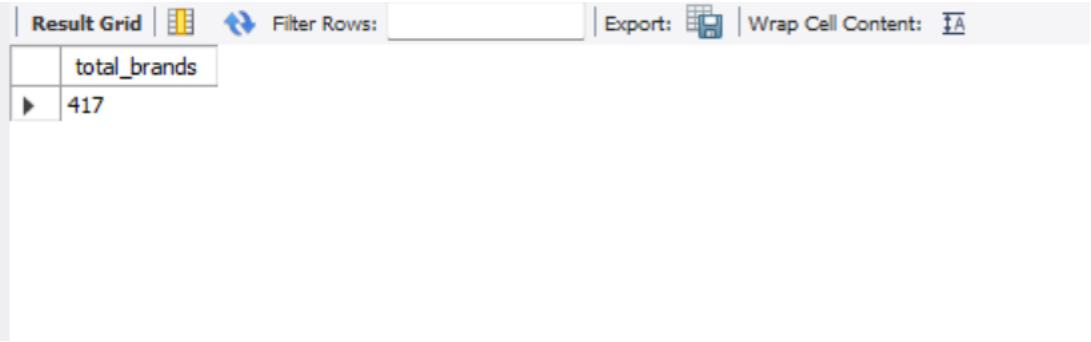
```
SELECT COUNT(*) AS total_products FROM myntra_products;
```



Insight: This query calculates the total number of product records available in the Myntra dataset. It gives an overview of the size of the product catalog.

2. **Find the total number of unique brands**

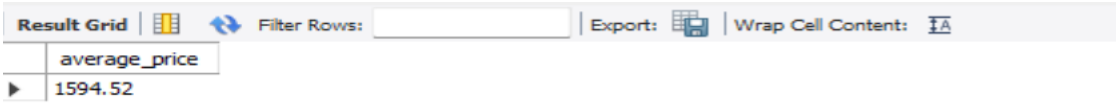
SELECT COUNT(DISTINCT brand_name) AS total_brands FROM myntra_products



Insight: This query identifies how many different brands are represented in the dataset. It helps understand the variety of brands available.

3. **Find the average selling price of all products**

SELECT ROUND(AVG(price), 2) AS average_price FROM myntra_products;



Insight: This query calculates the average selling price of all products. It provides an overall idea of the typical product price in the dataset.

4. **Find the minimum and maximum product price.**

```
SELECT MIN(price) AS minimum_price, MAX(price) AS maximum_price
FROM myntra_products;
```

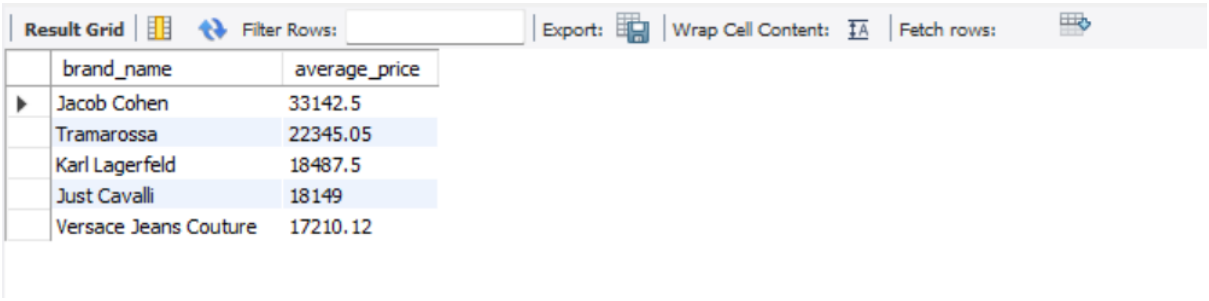


minimum_price	maximum_price
337	54000

Insight: This query identifies the cheapest and most expensive products in the dataset. It shows the overall price range of the products.

5. **Find the top 5 brands with the highest average product price**

```
SELECT brand_name, ROUND(AVG(price), 2) AS average_price FROM
myntra_products GROUP BY brand_name ORDER BY average_price DESC
LIMIT 5;
```

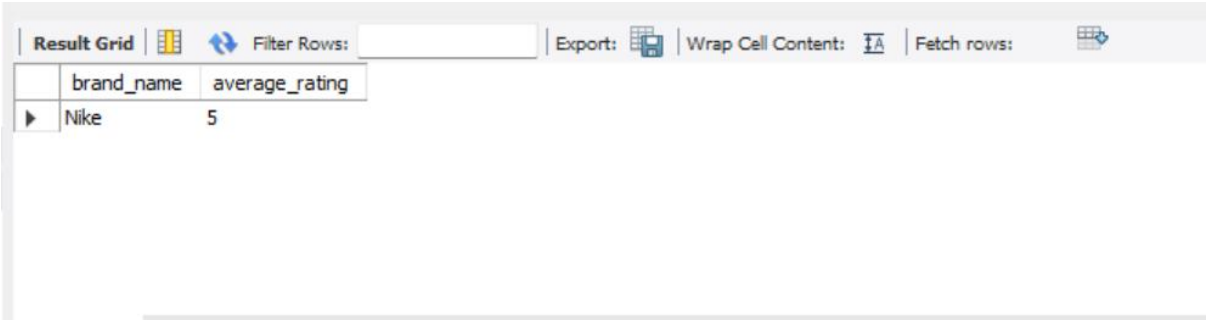


brand_name	average_price
Jacob Cohen	33142.5
Tramarossa	22345.05
Karl Lagerfeld	18487.5
Just Cavalli	18149
Versace Jeans Couture	17210.12

Insight: This query identifies the 5 brands with the highest average selling price, helping to highlight brands positioned at the higher-priced end of the product range.

6. **Find the brand with the highest average product rating**

```
SELECT brand_name,ROUND(AVG(ratings), 2) AS average_rating FROM myntra_products GROUP BY brand_name ORDER BY average_rating DESC LIMIT 1;
```



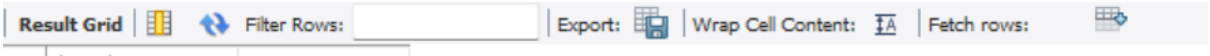
brand_name	average_rating
Nike	5

Insight: This query identifies the brand with the highest average customer rating. It can indicate which brand's products receive stronger customer evaluations.

7. Find brands having an average rating above 4

```
SELECT brand_name,ROUND(AVG(ratings), 2) AS average_rating FROM myntra_products GROUP BY brand_name HAVING AVG(ratings) > 4

ORDER BY average_rating DESC limit 5;
```



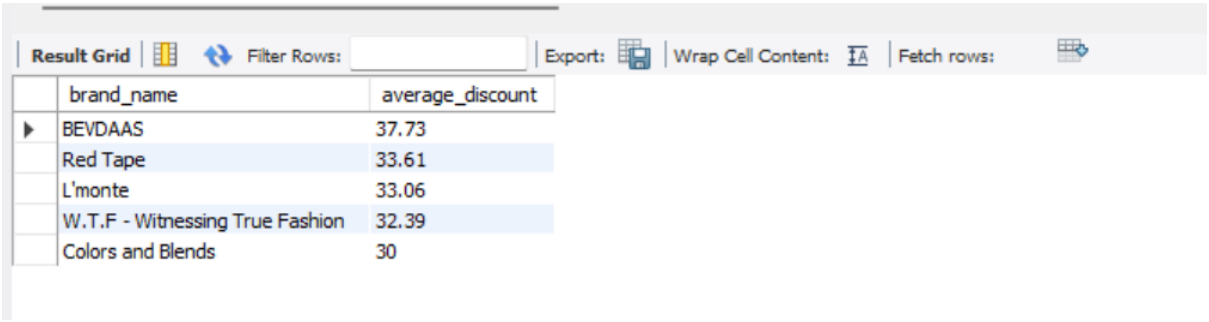
brand_name	average_rating
Nike	5
BAWSE.	4.8
Colors and Blends	4.6
Just Cavalli	4.6
Happy2r	4.6

Insight: This query filters brands whose average customer rating is above 4. It helps identify brands with consistently positive customer ratings.

8. Find the average discount offered by each brand

```
SELECT brand_name,ROUND(AVG(discount_percent),2) AS average_discount FROM myntra_products GROUP BY brand_name ORDER BY average_discount DESC
```

```
LIMIT 5;
```

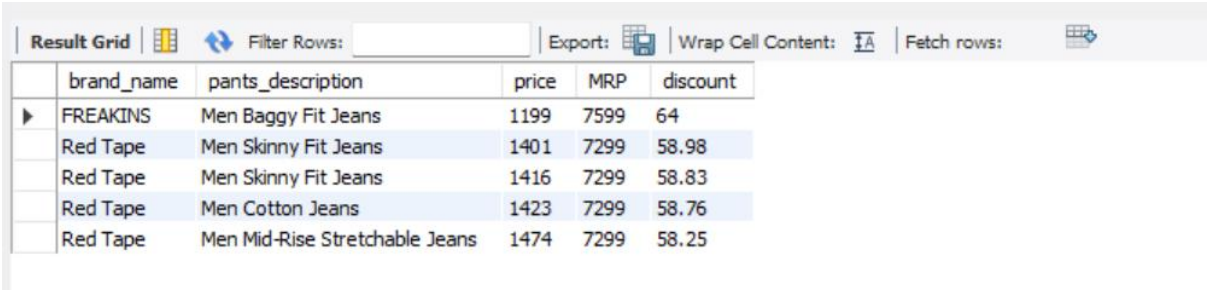


brand_name	average_discount
BEVDAAS	37.73
Red Tape	33.61
L'monte	33.06
W.T.F - Witnessing True Fashion	32.39
Colors and Blends	30

Insight: This query compares the average discounts offered by different brands. It helps identify brands that provide larger price reductions.

9. Find the top 5 products with a discount greater than 50%

```
SELECT brand_name,pants_description,price,MRP,
ROUND(discount_percent, 2) AS discount FROM myntra_products
WHERE discount_percent > 50 ORDER BY discount DESC
LIMIT 5;
```

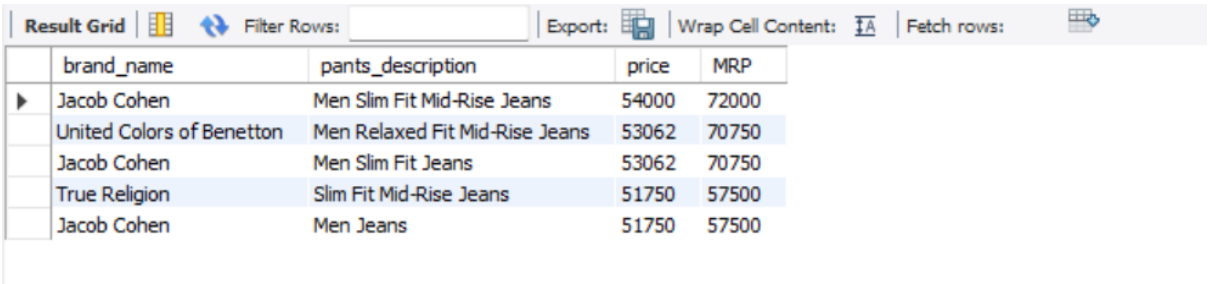


brand_name	pants_description	price	MRP	discount
FREAKINS	Men Baggy Fit Jeans	1199	7599	64
Red Tape	Men Skinny Fit Jeans	1401	7299	58.98
Red Tape	Men Skinny Fit Jeans	1416	7299	58.83
Red Tape	Men Cotton Jeans	1423	7299	58.76
Red Tape	Men Mid-Rise Stretchable Jeans	1474	7299	58.25

Insight: This query identifies products with discounts greater than 50%. It can help customers or businesses identify products with significant price reductions.

10. Find the 5 most expensive products

```
SELECT  brand_name,pants_description,price, MRP FROM myntra_products
ORDER BY price DESC LIMIT 5;
```

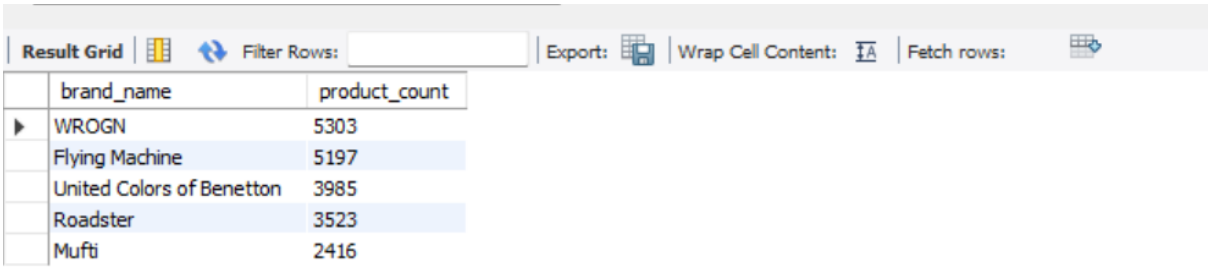


brand_name	pants_description	price	MRP
Jacob Cohen	Men Slim Fit Mid-Rise Jeans	54000	72000
United Colors of Benetton	Men Relaxed Fit Mid-Rise Jeans	53062	70750
Jacob Cohen	Men Slim Fit Jeans	53062	70750
True Religion	Slim Fit Mid-Rise Jeans	51750	57500
Jacob Cohen	Men Jeans	51750	57500

Insight: This query lists the five products with the highest selling prices. It helps identify the premium-priced products in the dataset.

11. Find brands having more than 100 products

```
SELECT brand_name,COUNT(*) AS product_count FROM myntra_products
GROUP BY brand_name HAVING COUNT(*) > 100 ORDER BY product_count
DESC;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

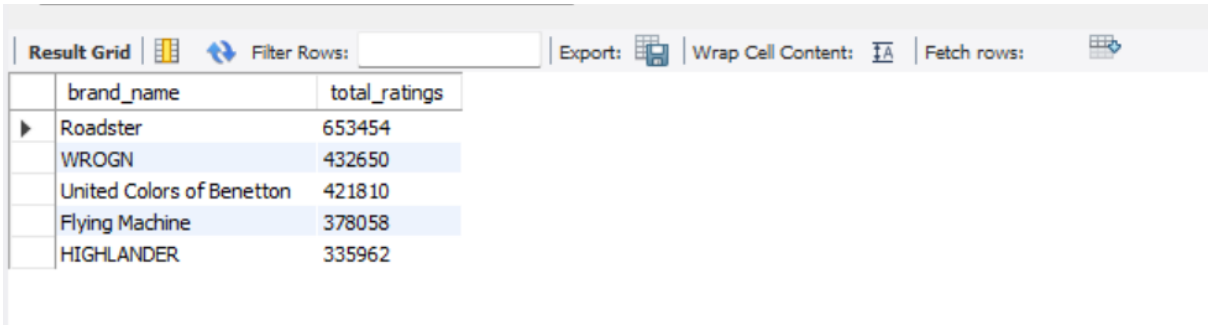
brand_name	product_count
WROGN	5303
Flying Machine	5197
United Colors of Benetton	3985
Roadster	3523
Mufti	2416

Insight: This query identifies brands with more than 100 products in the dataset. It helps determine which brands have a larger product presence.

12. Find the brands with the highest total number of ratings

```
SELECT brand_name,SUM(number_of_ratings) AS total_ratings FROM
myntra_products GROUP BY brand_name

ORDER BY total_ratings DESC limit 5;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

brand_name	total_ratings
Roadster	653454
WROGN	432650
United Colors of Benetton	421810
Flying Machine	378058
HIGHLANDER	335962

Insight: This query calculates the total number of customer ratings received by products belonging to each brand. It helps identify brands with greater customer engagement.

13. Find highly rated products with at least 100 ratings

```
SELECT brand_name,pants_description,ratings,number_of_ratings FROM
myntra_products WHERE ratings >= 4.5 AND number_of_ratings >= 100

ORDER BY ratings DESC, number_of_ratings DESC limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

brand_name	pants_description	ratings	number_of_ratings
Pepe Jeans	Men Stretchable Jeans	4.6	186
Flying Machine	Men Relaxed Fit Jeans	4.6	186
Pepe Jeans	Men Stretchable Jeans	4.6	186
John Pride	Plus Size Men Jogger Jeans	4.6	114
Flying Machine	Men Relaxed Fit Jeans	4.6	114

Insight: This query identifies products that have both a high customer rating and a substantial number of ratings. These products can be considered well-rated and relatively well-reviewed.

14. Compare discount levels with average ratings

```
SELECT CASE WHEN discount_percent < 20 THEN 'Low Discount'
WHEN discount_percent < 50 THEN 'Medium Discount 'ELSE 'High Discount'
END AS discount_category,COUNT(*) AS total_products, ROUND
(AVG(ratings), 2) AS average_rating FROM myntra_products GROUP BY
 discount_category ORDER BY average_rating DESC;
```

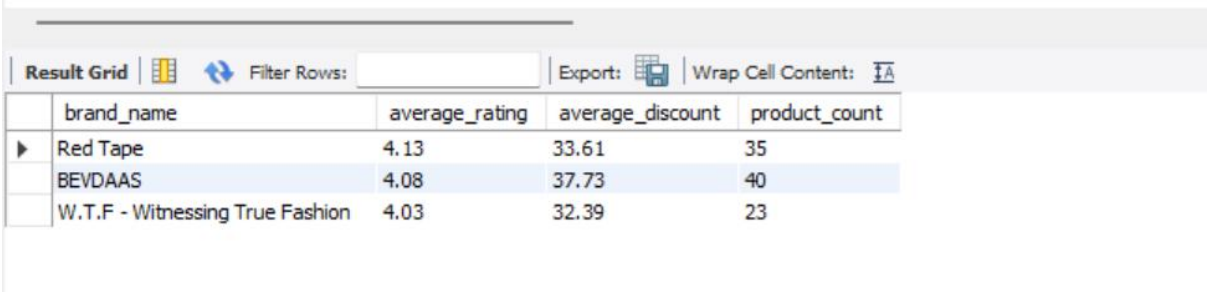
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

discount_category	total_products	average_rating
High Discount	28	4.1
Low Discount	50972	4
Medium Discount	1120	3.99

Insight: This query divides products into low-, medium-, and high-discount categories and compares their average ratings. It helps investigate whether products with different discount levels show differences in customer ratings.

15. Find brands with good ratings and high discounts

```
SELECT  brand_name,ROUND(AVG(ratings), 2) AS average_rating,
ROUND(AVG(discount_percent), 2) AS average_discount, COUNT(*) AS
product_count FROM myntra_products GROUP BY brand_name HAVING
AVG(ratings) >= 4 AND AVG(discount_percent) >= 30 AND COUNT(*) >= 10
ORDER BY average_rating DESC LIMIT 5;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

brand_name	average_rating	average_discount	product_count
Red Tape	4.13	33.61	35
BEVDAAS	4.08	37.73	40
W.T.F - Witnessing True Fashion	4.03	32.39	23

Insights: This query identifies brands that have an average rating of at least 4, an average discount of at least 30%, and at least 10 products. It provides a useful comparison between customer satisfaction, pricing offers, and product presence.

CONCLUSION

This project successfully analyzed the Myntra product dataset using MySQL to identify meaningful patterns in product pricing, brands, discounts, customer ratings, and customer engagement. Through SQL queries and aggregate functions such as COUNT(), AVG(), SUM(), MIN(), and MAX(), the analysis provided a clear understanding of the product catalog.

The use of GROUP BY, HAVING, ORDER BY, LIMIT, and conditional expressions helped compare different brands and identify products with higher prices, better ratings, significant discounts, and greater customer engagement. The analysis also showed how SQL can be used to transform a large dataset into useful business insights.

Overall, this project strengthened practical knowledge of MySQL and data analysis and demonstrated how SQL can be applied to real-world e-commerce data to support product comparison, pricing analysis, discount evaluation, and understanding customer preferences.